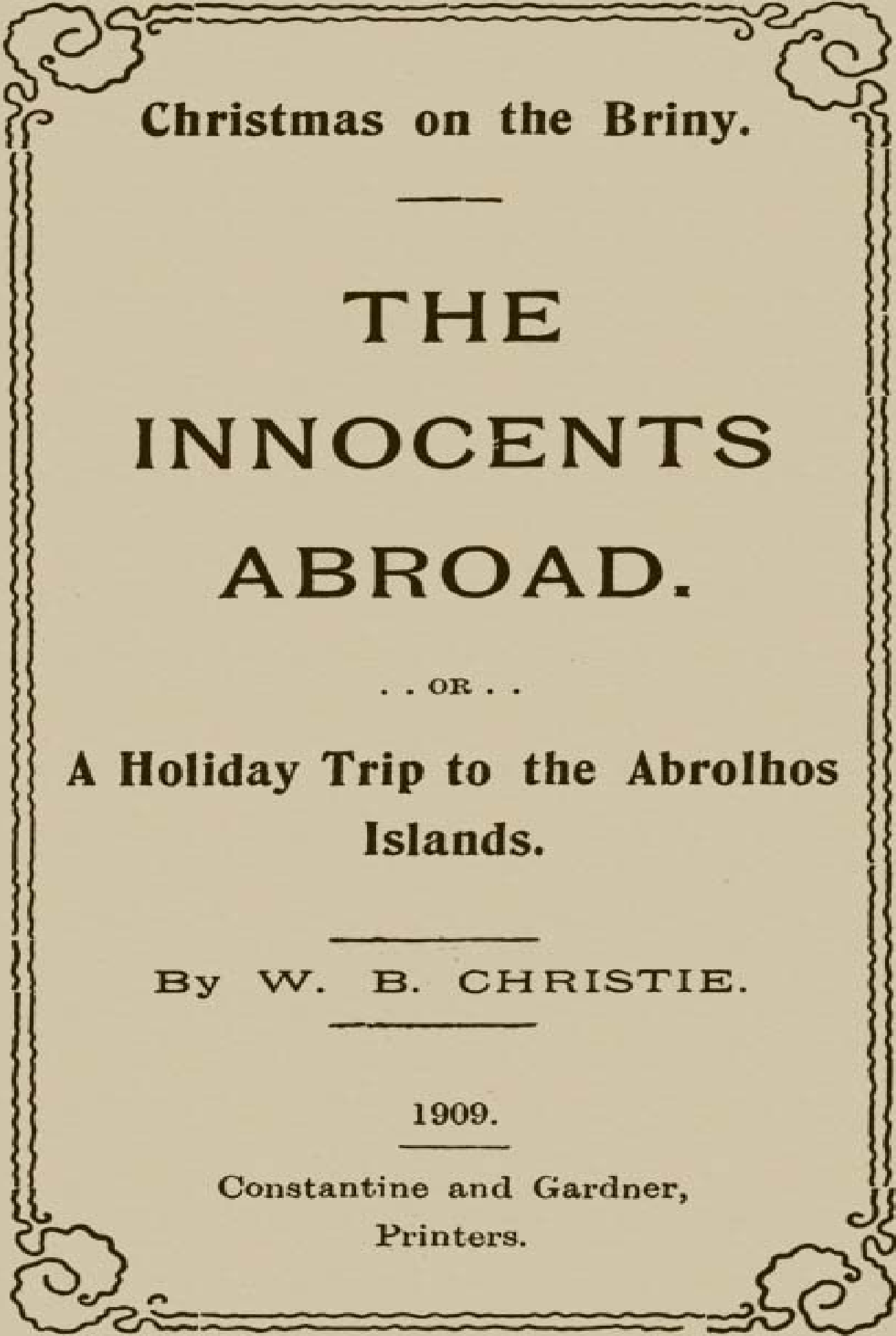




Christmas on the Briny.

**THE
INNOCENTS
ABROAD.**

... OR ...

**A Holiday Trip to the Abrolhos
Islands.**

By W. B. CHRISTIE.

1909.

**Constantine and Gardner,
Printers.**

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PUBLISHER'S NOTE.

The editor of the "Geraldton Guardian" has to acknowledge with thanks the author's courtesy in supplying him with the following interesting account of a trip to the Abrolhos Islands for publication in that paper. It has been suggested that their re-publication in booklet form might do much to advertise these islands as a holiday resort, and Mr. Christie courteously gave his permission, hence this unpretentious booklet.

THE INNOCENTS ABROAD.

CHRISTMAS ON THE BRINY.

(By W. B. Christie.)

“What shall we do with ourselves during the holidays?” was a question put by one to another amongst a dozen or so of the new and old residents of Geraldton a few weeks ago.

“What about the Abrolhos?” someone suggested. The idea caught on, and Mr. George Baston volunteered the use of his fishing boat the “While-away” to take a party of us for a cruise round the islands. The offer was accepted. The provisioning arrangements were left in the hands of Mr. Baston.

We were all to be aboard by midnight on Thursday, 24th December, but an hour before that time most of the party had staggered down singly or in groups of two or three to the Esplanade jetty, where the “While-away” was moored. Some were accompanied by friends to see the party safely off.

At twelve o’clock the cry of “A Merry Christmas” was reciprocated from shore to deck, and ere we had time to “blow the froth off,” the order was given to cast off the shore lines, and a few minutes later we were gliding slowly and silently through the maze of boats, which lay at their moorings, out into the expanse of still waters of Champion Bay, on whose face the gentle land breeze left scarce a ripple; out on to the mighty deep, on whose bosom some two or three of our party were to distinguish themselves as only landsmen can when they “go down to the sea in ships.”

We ought to have been in bed, but were so interested in the navigation of the Bay that our rugs were unrolled on the deck, and from the recumbent lounge of the hard boards we watched the leading lights of the Bluff as they

came into line, and, the helm being put down, we stood out through the channel in the Five Fathom bank, under the fitful and intermittent glare of the revolving light of Point Moore, which shot its rays far across the bounding billows of the ocean. But they didn't bound much; they simply rose and fell in long heavy undulations, and as our good boat climbed to the top of one, and gently slithered down, half sideways, into the trough beyond, some of our party crept softly to the side, and taking an apparent interest in the sparkling ripples as they danced past, remarked, "Ough-h-h," while another retorted "Ach-h-h"—remarks which were quite irrelevant to the general topic of conversation—about the memorable revolt of the shipwrecked crew of the Batavia on Pelsart Island, under the bloodthirsty "Captain General" 280 years ago, when the attempt of the Dutch to colonise New Holland came to an abrupt and tragic termination.

Point Moore light faded from view—the pleiades—Aldebaran, Orion and Sirius—had passed their meridian, and were slowly sinking to the West. Achernar had dipped into the bank of clouds which lay over the southern horizon, Canopus blazed and twinkled as he swept in his majestic circle round the pole; the Cross and its attendant Centauro were clinging to their upper culmination, when the grey of the east told us that Christmas morning was about to break.

In the gathering grey the stars faded where they hung, and as the light broadened, we saw that our horizon was bounded by the heaving waters. The fleecy and burnished clouds seemed to part to make way for the sun as he emerged from the eastern sea and lighted up a glittering path direct to our craft, as though he had singled us out as the only participants of his glory. Scarcely a ripple was on the water. The sails flopped lazily, and with every heave of the swell the boom gave a dull thud as it jerked the main sheet taut, and lazily swung back to gather force for the next thud.

"Coffee. Who says coffee?" shouted the chef, who was presiding at the primus stove below. We all said "coffee," and a mug of that steaming beverage, with a biscuit, was passed round to all hands as we lay in our rugs about the deck.

"How's her head?" someone called to our skipper at the helm.

"Her head's anyhow," growled Nicholas; "we're becalmed," and the boom gave another thud as we slipped over the summit of the swell.

“What depth of water here?”

“About twenty fathoms,” replied Nick.

“Any fishing here?”

“No; it’s a sandy bottom. Try if you like.”

Lines were got out, baited and thrown over. How very true the description of fishing: “a long string, with a hook at one end and a fool at the other.” Not a bite rewarded our efforts, although we sat with the lines till long after breakfast. We threw over an empty match box. It bobbed up and down beside us for a long time till a little puff of wind blew us away from it, and then ceased so that the boom might continue its flopping, as it toyed with the main-sheet. The conversation drifted from the Captain General to the facility with which mutinies could be fomented on sailing ships when they were becalmed. Then, naturally, from mutiny to slavery, and the condition of Carolina was discussed, and the wisdom of the remark made by the Governor of the northern half of that State to the Governor of the southern half—that it “was a long time between drinks.” The idea caught on like a fish-hook to a trouser leg, but it took Mr. Baston, who was in charge of the locker, some time to find the corkscrew. But when he did find it—ah!

Fitful little puffs of wind tightened the sails at fitful intervals, and sent the waters sparkling behind us, as the sun began to sink towards the west. As sundown approached, a steady but light breeze began to waft us slowly forward. The light clouds of the western horizon were painted in bright golden hues, and the sun sank beneath the waters a glaring ball of fire amid the living flame into which he had touched the burnished clouds which overhung him.

“Plenty of wind by and bye,” remarked the skipper, as our sympathies again went out to the slaves of Carolina. Under the influence of the rising breeze our boat danced merrily through the waves until Nicholas, who had mounted the masthead, said he could hear the distant roll of the breakers as they broke over the coral-capped reefs of the Abrolhos, and we must lay the boat to for the night. The boat was thrown up into the wind, and everything made safe as the new moon sank beneath the horizon, and under the splashing music of the waters we rolled into our rugs and were lulled into that soft slumber which was only disturbed by the hardness of the deck on

which we lay. An occasional “Ough—h—h” or “Ach—h—” smote dreamily on our ears, telling us that some, at least, of our party did not trouble particularly whether the ducks came home or the cows laid.

Thus was our Christmas spent. “Coffee!” shouted Miles at about five o’clock, and we roused up to find the boat slightly careening under a pleasant breeze, which was blowing us along at about six or seven knots. We had drifted a good way back and northerly during the night, and it was not till near ten o’clock, before we sighted Goss Island, the most easterly of the middle group of the Abrolhos—a low, sandy spit, rising only a few feet above the surrounding waters. Our unaccustomed eyes could not distinguish it from the crested foam of a rolling wave till long after our watchful skipper pointed it out to us, and told us he was on the look-out for a beacon-pole which was erected on its shore.

Passing this island at a respectful distance, so as to keep well away from its outlying reefs, the Wallaby and Pigeon Islands hove in view, and, entering the channel which lay between them, sailed up to a sheltered cove, which afforded good anchorage, and dropped anchor at about noon, having been thirty-six on a journey which with a moderate breeze would have occupied about eight.

PIGEON AND WALLABY ISLANDS.

The channel narrowed till it was not more than three hundred yards or so wide. Pigeon Island lay on our left, with deep water right up to its precipitous rocky bank. The island is one rocky coral mass, with scarce a sign of vegetation. Its surface is covered with mounds of rocks thrown together or built into rough walls gleaming white in the sun, and we are told that these stones were built up as they were removed from the deposit of guano which was worked here. Sixteen hundred tons of this useful manure has been shipped from the island, which has an area of perhaps 200 acres. Going ashore we find it a barren, inhospitable rock—a gull or mollyhawk sits here or there—a few hover about, but there is no sign of the bird life which we were prepared to see here in such abundance. Since the removal of the guano, the island appears to have been so disturbed that the birds have almost entirely forsaken it. It is too rough to walk about on with any comfort, so, looking round it, we return to the dinghy and put off to the boat, where luncheon is ready for us.

To the right, and about half a mile away, is East Wallaby Island. From the deck we see the channel line clearly defined against the shoal water, out of which the island rises. As four of our party have to get back by Monday to resume their business—and this is Saturday—some of those who are going to remain put off in the afternoon to Wallaby Island to select a camp in which to wait till the boat returns from Geraldton, whither she is to start on Sunday morning. We expect her to return by Tuesday, at furthest. There is little likelihood of rain, and we are going to camp on the beach, with the prospect of plenty of sport in fishing and wallaby and pigeon shooting. The dinghy puts off with four of the party, her full capacity, but ere she has reached halfway to the shore the water suddenly shoals and they have to get out and walk the remaining distance up to their knees in water, over the sharp coral rocks which form the base on which the island stands. It is fortunate that we brought strong, heavy boots with us, for the sharp coral would have cut light ones to pieces, and it would be impossible to walk over that stretch of shallow water in the bare feet. The prospecting party returned in a couple of hours with a couple of wallabies, and reported

having found a good wind-break, where Mr. Drewry had camped some twelve months or so ago. In the afternoon the dinghy was again despatched with provisions, cooking utensils, and a sail to cover over the wind-break, and so provide a shade shelter for the eatables. Those left on board the boat put out the lines and in a very short time had a good bucketful of small schnapper. Mr. Randell proved himself to be the piscatorialist-in-chief, and to the culinary skill of Mr. Nathan we owed a delightfully fried fish supper.

In the morning wading operations had to be renewed, while breakfast was being prepared, and bedding and water, were transported to the camp. Our case of beer was sensibly diminishing. The Governor of North Carolina would have his say, and, amongst other things which we were to give was an order to the boat to bring out from Geraldton, was another case, so as to give "His Excellency," as someone remarked, a fair show. The transportation of our party and bedding, however, took two trips of the dinghy—and in the hurry of our departure, the order for the beer was forgotten, and not thought of till the boat had hoisted sail and was fairly under way. One of the legal members of our party thought he could convey the message by signs, and, holding up a bottle and shouting frantically, he ran along the shoal water towards the boat, when, tripping over a sharp piece of coral, he fell headlong with a splash into a deeper hole, amid roars of laughter from boat and shore, in which further opportunity for conveying the message to the fast receding boat was lost. We sadly realised that, if not absolute teetotallers, we would at least have to place ourselves on short commons, as far as beer was concerned, for the remainder of the trip. Water which has been stored for some days in a boat's tanks is not the pleasantest of drinking. There were eight days of the trip yet to do, and only eighteen bottles of beer and four of whisky amongst six of us. Mr. Baston was placed in charge of the liquor, while one other learned member of the Bar was placed in possession of our only corkscrew, with strict injunctions that neither was to listen to the syren voice of the Governor of Carolina, except twice a day, and then only when every member of the party was present—an injunction which it is gratifying to note, they both rigidly adhered to.

Wallaby Island is about three miles long by a little over a mile wide, and is the hilliest of the group. Sand-hills have been blown to a height of about 50 feet, and cover a good portion of the bare coral rocks, of which the island is composed. The sand is mixed to some extent with guano, and for the most

part the island is well clothed with short scrubby vegetation. About a quarter of a mile north of our camp, which we named "Point Desolation," a mast has been erected on the summit of the highest hill as a beacon to navigators of the intricate channels in the neighborhood. Some 200 yards north-east of this beacon, on bare limestone rocks, is a natural well, or gnamma hole, about ten feet deep, with a good supply of drinkable water. It is well for tourists to know of the existence of this well, as otherwise the islands appear to be devoid of fresh water.

Wallabies are fairly numerous on this island, but as the island is of very limited extent, they should be protected from indiscriminate destruction, except for food. A party with a few guns would exterminate them all in a week or two. Pigeons also are fairly plentiful. Most of these islands are merely banks of dead coral elevated only a few feet above the sea and devoid of vegetation, and on those few exceptions which are capable of supporting animal life, the game at least should be preserved as a possible means of life to shipwrecked or weather-bound crews who may be compelled to remain on them for some time. The southern shore of the island is, for the most part, bare and rocky bluff headland, but on the northern side there is a pleasant beach extending up to Turtle Bay, the only really pretty and pleasant spot on the whole island—or indeed on any which we visited. There the sandy beach shelves gradually out into deep water, and while there we daily walked the three miles along the beach to the bay to bathe, and spend most of the day. Plenty of sponges, known as the abdominal variety, lie about on the beach. They are, of course, dried, and for the most part rotted by the weather and exposure, but some are of fine texture, and appear to show that the waters may be worth prospecting with a view to opening up the industry. The pretty little Lesser Tern is numerous among the bird life. She lays a brown spotted egg in exposed situations on the bare rock. In the season they are said to possess a delicate flavor. Some hundreds were gathered.

Tuesday evening arrived, and no sail was visible from our look-out on the hill. Our last loaf was cut into for supper, and, unless the boat arrived soon, we would have to bake a damper, but the difficulty was in getting wood

enough for the fire to bake it in. We, however, had plenty of potatoes and onions, and crayfish were plentiful, so we were in no fear of starvation, but it was disappointing for the boat not to have showed up. With a few pigeons shot by Mr. Kidd and curried crayfish and vegetables, we regaled ourselves, and washing our supper down with our evening modicum of beer, turned in on the sands and talked till we fell asleep. The wallabies, unaccustomed to seeing men or dogs, are exceedingly tame, and during the night hopped about the camp within a few yards of our beds.

RAT AND WOODED ISLAND.

The Post Office—Weather bound—Home.

Wherever one looks from Wallaby Island, little rocky islets stand out from the coral plateau, and many of them can be waded to. West Wallaby is another good-sized island a mile and a half or so from our camp. There is, however, a deep fissure-like channel between it and us, and the distance is somewhat great to wade on the sharp, coral bottom which intervenes. At low tide this rocky plateau is almost level with the water, but we found plenty of oysters on the islets we visited.

On Wednesday we looked anxiously for the boat. Our stay on the island was becoming monotonous, as we had no means of leaving it except where we could wade to. We knew that plenty of fresh bread would be on board, and some other supplies which we found we required, as well as water, of which we were running short. About three o'clock, just as we were giving up hope of her arrival that day, a sail was descried coming over the horizon, and, in an hour or so more, all doubts of identity were dispelled as the white hull of the "While-away" hove in sight, and she bore down to her anchorage at Pigeon Island.

Although we had had a good steady breeze during the four days of our stay on Wallaby Island, the boat had become becalmed on her way to Geraldton, and the passage across had taken some forty-eight hours. The tide being high, we at once struck camp, and shifted our chattels aboard that evening so as to be ready for an early start to Rat Island, some 14 miles to the south, next morning.

Rat Island is probably the largest of the group, and is a level plateau of coral standing about ten feet out of the sea, with an area of about 500 acres. Here extensive deposits of guano have been worked, and a large stone shed for sorting in, besides huts and tram lines, were erected, but the island having been worked out, everything worth taking away has been removed to the present workings at the Pelsart group. Only the walls of the buildings now

remain. What was once a good stone jetty runs a few chains out into fairly deep water. A few Italian fishermen reside here, and, with their half-dozen boats moored near the jetty, give the place a comparatively busy aspect. We found them most obliging in every way. As we approached, they put off in their dinghy and assisted us to our moorings at the head of the jetty. After mooring, we went ashore to prospect.

The island is well clothed with low shrubs, and in many parts there is a dense sward of wild oats and silver grass, while numerous patches of ice-plant denote that the soil is rich in nitrogenous matters. About a hundred acres of soil could be got on this island which would grow prolific root crops.

What attracted our interest most was the sea fowl. Millions upon millions of sooty and noddy terns rose in clouds, and circling round for a few minutes, settled again. This is one of the breeding grounds, and hastening over to the place, the birds rose in clouds at our approach, and circling round us, almost within reach of our hands, resented our intrusion by their deafening cries. Every shrub and bush had from two to half a dozen nests on it, all with one egg, on which the hen birds were sitting. On our approach the bird would rise, but in a few minutes would circle round to the nest again, and settle on the egg within three feet of us. Thousands were already hatched, and in all stages of development, and we had to pick our steps to avoid treading on them. The old birds show a strong parental instinct, and resent any interference with their young by cries and savage pecks, but with a little patience they soon become quiet, and will sit beside the young while you place your hand on them. Here and there on clear patches thousands of the young in all stages of growth are gathered together, covering the whole ground, while a few of the older birds hover about them. It looks like a densely packed feathered school, with their teachers in charge. Every here and there are these schools, while on and under every bush there are stragglers apparently playing truant. The noddy terns are about the size of pigeons, with rich chocolate plumage. They take possession of the bushes, while the sooty terns seem to lay their eggs on the bare ground without any attempt at a nest. In parts the ground is so littered with eggs that a foot can hardly be put down without treading on them. They mix freely amongst themselves, and as they rise in flight, every few minutes, they can be

compared to nothing better than an immense swarm of bees, covering acres in extent, while their deafening screeches drown every other sound.

In the evening we found the beach and rocks about the jetty literally alive with crayfish, so, baiting one of our fish pots, we threw it over the side, and before bed time had our dinghy half full of fine crustaceans. The young schnapper and whiting bit freely, and we had a good catch, Mr. Randell maintaining his reputation as a piscatorialist.

New Year's morning broke fine and clear, and most of us spent the forenoon in further explorations of the island, while our skipper and chef took advantage of the Italians' hospitable offer of their wood fire on which to boil a dozen or so crayfish. Before noon we bade our friends good-bye and stood out down Zeewych Channel towards Woody Island, some ten miles south. This island seems to derive its name from the fact that two mangrove trees grow in a salt lagoon on it. That is the only sign of wood which it presents. There are a few low shrubs, but on the whole vegetation is scanty. Here considerable quantities of guano have been secured, but the island is now worked out. The island is cut into two by a narrow patch of shoal water, through which we can wade at low tide. Bird life here is almost as numerous as at Rat Island. The western half of Woody Island is a ring of dead coral surrounding a landlocked central lagoon of salt water. The sooty terns' eggs are scattered all about the margin of this lagoon, and appear to be hatched by the sun's heat, as we saw none of the birds sitting on the eggs. Almost every limb of the mangrove trees has its noddy tern, and the young are getting about, though unable to fly. On the other half of the island the birds rise in millions, forming a black circling cloud against the sky. The lagoon, though mostly shallow, is in some parts very deep. As we stand on the rocky bank wondering how this could have been formed, without any visible connection with the sea, a turtle of some 300lbs. weight happens slowly along, and raises its head to reconnoitre. On seeing us, he immediately dived and scuttled off for the deep water with astonishing velocity for so ungainly a form.

Our boat was moored to one of the piles which once formed the end of a wooden jetty erected here. We put off to her in the dinghy in time for tea. Some threw out the lines, but Nicholas told us there was no hope of fish there. We did not get a bite.

The wind rose steadily, and by the morning it was blowing a stiff gale, so taking in a reef on the mainsail, we set out for the Pelsart group, some 15 or 20 miles further south. The wind, however, increased, and it was thought advisable to run for shelter into a channel between two bare coral islets near the “Post Office.” Why it is called the “Post Office” it is difficult to understand, for no one lives within miles. On the extremity of one of the islets a beacon of stone has been built, about the size of a sentry box, and it is said that this gave rise to the name. Names are evidently very easily suggested.

We had to beat in to our shelter. The channel is not more than a hundred yards wide, and flanked on either hand by sharp coral rocks, which voraciously bite a hole in the bottom of our boat if we came into contact with them. The islets are forbidding in their aspect, and although oysters are to be had on them, the difficulty of walking over the loose, sharp corals make us glad to return to the boat, where we may get below into shelter from the now howling gale. The anchor drags a few yards now and then over the rocky bottom, and a second is got ready to throw over, but happily the first at last catches a good hold, and keeps us in safety from the fury of the wind.

Square Island, Wreck Point, “Batavia’s Grave,” and Pelsart Island are visible in the distance to the south, but the living gale keeps us weather-bound at our moorings. All Saturday night it howled and whistled through the rigging.

Poor old “Father,” who had hitherto insisted on sleeping on deck, realised with Sir Joseph Porter, in “Pinafore”—

“When the breezes blow I generally go below,
And seek the seclusion of my cabin grants,
And so do his sisters and his cousins and his aunts.”[Pg 27]

The gale was too much for “Father,” so he joined the others in the hold, as our vessel writhed under the fury of the blast and tugged at her moorings.

Sunday morning broke with the wind unabated. The long, spray-capped ocean rollers could be seen over the top of Pelsart Island, as they thundered over the reefs and churned themselves into seething masses of foam. There was no hope of shifting that day. Nicholas told us it would be madness to

attempt to run to Geraldton in the heavy sea which had been lashed up by the southerly gale, and it was hopeless to attempt to beat against that gale to Pelsart or the other historical places we wished to visit. Our bread was run out, and we made puff-de-lunes, but the rolling of the boat was not conducive to the culinary art. The biscuit tin was nearing the bottom, and we were reduced to one biscuit a meal, although we had plenty of tinned meats, and fish and some oysters. The last two bottles of beer were opened as the sun approached the yard-arm, and were skilfully divided amongst the eight of us.

Towards evening the wind lulled, and the sky became overcast, so it was decided that as soon as practicable, we would start for home. Nicholas told us the glass was falling again, and someone remarked that the "Governor of North Carolina" had been finally deposed at 11 o'clock.

By Monday morning the wind had considerably abated, and the sea had gone down a good deal, so that by 10 o'clock it was pronounced safe to face homeward. The anchor was hove, and in a few minutes we were bowling along homeward under a still fresh gale. Showers of spray dashed against the boat and sent most of us below. As we approached the open sea, we found it still running quite as high as we wanted it. It was too rough to think of cooking anything, so dividing a Swallow and Ariel plum pudding amongst us, "Father" and a learned member of the Bar retired to the seclusion of the hold to assiduously cultivate a thirst, which they thought would arrive at maturity by the time we reached Geraldton.

At noon, Wizard Peak, Mt. Fairfax, and the Moresby Range hove in view, and by three o'clock our chef shouted down that the lighthouse was visible, and we would be at anchor by half-past four.

Alas for the uncertainties of wind-jamming craft! The breeze which had been gradually dying, forsook us altogether, and we lay helpless, but not motionless, on the heavy billows. We had ample time now to study the geography of the coast line from a distance as we rose and fell from the trough to the crest of the seas. The western horizon was a sea of flame as the sun sank beneath the waves; the stars came out and twinkled mockingly

at us to the music of the thudding boom as it jerked the main sheet to the end of the horse; steerage way was lost, and we flopped and floundered round all points of the compass. For seven weary hours we watched the shore. Point Moore light blazed on us in its intermittent flashes, the lights of the town shot their rays across the waters to us as we picked out those of well-known hostelries with which we were so familiar. They only accentuated a thirst which had already arrived at its full maturity, and it was not till near eleven that at last we dropped anchor, and came ashore at the pier, blistered, sunburnt, storm-tossed, but full of life and hope.

Never were higher encomiums passed on the excellent management of the Globe Brewery! Never were more unstinted praise to the tasteful skill of the brewer, as we performed the final act of a ten days' most enjoyable outing.



Constantine & Gardner, Printers, Geraldton.

The Abrolhos Islands.

INFLUENCE OF OCEAN CURRENTS.

THE ZOOLOGICAL PROVINCE OF AUSTRALIA.

Although there are few places on the Australian coast where one can spend a more enjoyable summer holiday than the Abrolhos, both on account of the free and “simple life” they afford, and the great historic interest they possess as being the scene of the final catastrophe which befel the Dutch attempt to colonise Australia in 1629 under the unfortunate Pelsart, the islands themselves are of peculiar interest, and have given rise to much speculation as to their origin.

They are all composed entirely of dead coral, here and there partially covered by wind-blown sand, and for the most part not rising more than ten or twelve feet above the water, while many of them do not attain an elevation of more than three or four feet. They are situated on the 29th parallel of south latitude, and are probably further removed from the Equator than any other coral islands in the world, while the indigenous vegetation, such as it is, belongs to a latitude well within the tropics. It is a curious speculation therefore, how these islands came to exist in their present position.

When we “lay to” on our recent trip our boat drifted considerably northward, while closer to Geraldton the northerly current is still more defined. These northerly flowing waters are the cold waters of the Southern Ocean travelling into the equatorial regions, where they are warmed and supply the current which passes westerly and southerly through the Indian Ocean, attaining its greatest velocity as it passes through the Mozambique Channel, thence round the Cape of Good Hope, and north-westerly through

the Carribean Sea to form the Gulf stream, which takes a north-easterly course to impart their warm, genial climate to the British islands.

The equatorial current of the Pacific is broken up and delayed in its course as it passes through the Indian Archipelago and Sunda Islands, and its waters become warmer than those of any other oceanic waters on the globe, while the current itself is split up and diverted into innumerable directions, which make navigation extremely intricate and dangerous. It finally emerges from this intricate maze of islands at about latitude 15 south and longitude 115 east, where portion of its waters join the easterly current to Mozambique, and the remainder flows about south by west, being kept from the Australian shores by the stream of cool water from the Southern Ocean, which hugs the Westralian coast line till it reaches well into the tropics, and merges into the warmer water of Oceania between Sharks Bay and Java, as a lower current. A glance at the map of the world will show exactly how the islands of the Indian Archipelago would operate in forming the intricacies of the currents along the north-west coast.

Owing to the rotation of the earth being from west to east, the tendency of all currents in the ocean is to flow westward, unless some local cause deflects them from that course. The most notable example of this deflection is the Gulf stream, while the next most important is the Pacific current in its ramifications through the islands of the Indian Archipelago.

Polar waters always flow towards the equator, still partaking, however, of the westerly direction unless deflected by local causes; and, being cold, flow beneath the warmer currents, which have been expanded by equatorial heat, until, being warmed, they gradually rise and mingle with the equatorial currents, which, in addition to their westerly motion, also flow towards the Poles, till, becoming cooled, they sink and merge with the Polar waters, thus maintaining a continued circulation.

Thus the polar current which flows northerly past Geraldton has come up from the south of Australia, and being deflected by the southern currents from the Pacific and Indian Ocean, has been pressed, as it were, against the coast line, attaining its narrowest part as it passes beyond Sharks Bay.

The longitude of the Abrolhos Islands is about 113-30E., and here we may refer to a bottle which was picked up on December 9th—two or three weeks before our visit—on East Wallaby Island, by Mr. F. Burton, which has an

important bearing on the question under discussion. This bottle was thrown over from the German ship Innsbruck on the 2nd June in longitude 111-41 E. and latitude 21 S., and, as mentioned in the "Guardian" of 15th December, which quoted the written paper it contained, was for the purpose of determining the course of the current from that point. We do not know how long this bottle was on the beach before it was discovered, but we may be pretty certain that the longest portion of its period afloat would be after it arrived at the Abrolhos, in battling with the local counter currents which the islands themselves would produce. During the six months which elapsed from the time it left the Innsbruck, it had travelled through 8 degrees of latitude southward and 1-11 of longitude westward—a distance of about 600 miles—in nautical phrase, about south by quarter west. The position 111-41 E. and 21 south would be about 80 miles west of Flaming Head, and although we have no data on which to estimate the rate of the current, the bottle gives us absolute information as to its direction—south by quarter west.

Australia, and the whole of what is known as the zoological province of Australia, stands on an ocean plateau which rarely exceeds 100 fathoms in depth; while immediately beyond it, the ocean suddenly attains a depth of from 1000 to 1,200 fathoms. The edge of this plateau is at Lombock Strait, between Lombock and Sambawa Islands, to the east of Java, and passes northerly through the Strait of Macassar, between Celebes and Borneo, thence easterly, embracing the Molucca Islands and New Guinea, and southerly between Australia and New Zealand and embracing Tasmania. The Abrolhos also stand on this plateau.

Thus all the Pacific waters which pass to the eastward of Celebes and Lombock pass over a considerable portion of this plateau, where the water is only 100 fathoms deep, and therefore absorb a much larger amount of equatorial heat than the waters to the west of that line do, where they attain a depth of 1000 to 1,200 fathoms.

This eastern water, of the Australian province, heated by passing over the Australian plateau, is the water into which the captain of the Innsbruck threw his bottle. Thus, while the climate of Geraldton is tempered by the cool waters of the northern current from the Antarctic Ocean, the Abrolhos have an undue share of warmth transmitted to them by the heated waters of the Pacific as they enter the Indian Ocean. It is this undue share of heat

which places these islands in the unique position of being the most southerly coral formations of the world. The coral insect can live and work in the temperature which the waters of the Pacific convey to them to a latitude beyond which the same temperature is denied elsewhere.

Thus most of the flora and fauna of these islands are distinctly tropical in their character, although their habitat is situated six degrees outside the tropics.

The Australian plateau is not a mere geographical phenomenon. It is more, and forms the boundary between the zoological provinces of Australia and Malay. To the east of Lombok Strait everything—birds, animals, fish and even molluscs—are distinctly Australian in their types. To the west they are as distinctly Malayan, and although Lombok Strait is less than twenty miles in width, yet the flora and fauna on either side of it are as distinct as though they were separated by half the diameter of the globe. The birds, which could easily bridge the distance in their flights, or the fish, which could traverse the waters in an hour, remain distinct and faithful to the type of their own zoological province. In the flora, too, the trees and shrubs, the grasses and the herbs, true to their province, retain their botanical individuality, with as much persistence as though separated from their neighbors across the strait, by the broad waters of a mighty ocean and give a contradiction to the doctrines of Darwinism that migration has anything to do with the origin of species, or that natural selection plays any part in permanency of races.

The whole surroundings go to prove the truth of the observations of Agassiz—that each zoological province has brought forth life best suited to its own chemical and physical environment, without reference to the life already existing in the neighboring provinces.

THEIR PHYSICAL CONDITION.

A SHRINKING WORLD.

Another point which forcibly strikes an observant visitor to the Abrolhos, is that the mass of most of the islands is composed of dead coral with scarce a trace of soil except when the guano deposits have, in a few localities, become mixed with the wind-blown sand, which has enriched such patches so that they now grow stunted shrubs, and on the richer of these patches a species of wild oats, and silver grass, both of which have evidently been introduced. This is specially noticeable on Rat Island, where particularly rich patches aggregating probably 100 acres, grow a most luxuriant and dense crop of these grasses. But Rat Island was for a long time worked for its guano deposits, tramways were built and horses were stationed there, so that the seeds brought over with their fodder have germinated in the better portions of the island. These portions would thus at once become the favorite feeding grounds for the horses, whose droppings have further enriched the land, till now it would be capable of cultivation with good results.

But it is the islands in their natural state, such as the Wallabies, Pigeon Island, Woody Island, and the numerous islets which rise from three or four to ten or twelve feet above the sea—all coming under the description of “dead coral”—which attract attention. The coral insect does not build above water, nor does it build at all in water beyond a moderate depth. How, then, do these coral islands come to exist in their present position?

There are two possible explanations for the existence of these islands. Either the plateau on which they stand has been upheaved, and thus pushed them up from below, or the ocean itself has receded, and left them exposed to the atmosphere. Either of these would have caused the destruction of the insects who build them. A review of the evidence in support of these causes will be of interest.

For land to be upheaved it must necessarily be removed to a greater distance from the earth's centre than it was previously, and there are only two kinds of forces which can effect such a removal. First, the recoil of

volcanic action at a distance. When we discharge a gun, half the force of the charge is expended in driving the bullet, and the other half in its reaction against the shoulder. If we push a weight, half the strength exerted is in moving the weight, while the other half is spent in holding our feet in position. And so with all forces—the amount spent in accomplishing its object is equalled by the resistance against which the force is acting. Thus, if a volcanic effort were being made at, say Karakatoa, which is the nearest volcanic vent to the Abrolhos, the force expended in the outbreak would be equalled by a recoil acting on some part of the viscid interior of the earth, in the opposite direction to the seat of force. So that if the seat of force were to lie between Karakatoa and the Abrolhos, one portion of the force would have its visible effect at Karakatoa, while the other might drive the molten interior backward beneath those islands, and thus elevate them. But in any case, by the displacement of matter at the seat of force, a vacuum would be caused which, in its turn, would be compensated for by the subsidence of a portion of the earth's surface, or ocean bed, to make up for it.

As a rule, these subsidences are sudden, and, if they take place on land, are always disastrous. Thus the great earthquake at Lisbon in 1775, was so sudden that in less than six minutes the land had sunk 600ft., and of 60,000 persons assembled on the new marble pier, not one of the bodies ever rose to the surface. At the same time a town in Morocco, with 10,000 inhabitants was swallowed, and not one escaped. At Kingston harbor, in Jamaica, a similar subsidence took place with equal suddenness, and for nearly a century afterwards the remains of warehouses could be seen a hundred feet or so below the shipping. And so on. Instances may be multiplied. The Runn of Cutch in India, the sinking of the South Island of New Zealand, the disaster at San Francisco, and the catastrophe of Messina, all are the effects of the same forces which result in the sudden alteration of relative levels.

There is another force which also causes upheaval. If two subsidences take place, causing what are known as “deeps” in the ocean, in proximity to each other, they act as wedges on the ocean bed intervening, and force it upward, often folding and distorting stratified deposits, in a remarkable manner, but always forcing it into ridges and mountains which frequently are volcanic. One of the most notable instances of this kind of upheaval is New Zealand. On each side of it is a “deep,” caused by subsidence of the ocean bed. These acting against each other, forced New Zealand up into high ranges, which at

once became the theatre of volcanic action. One of these “deeps” lies to the east of New South Wales.

Another instance is the “deep” which lies to the south of, and is the cause of, the Australian Bight. This “deep,” acting against the “deep” to the east of New South Wales, forced up the south-eastern corner of Australia, and, if any part of Australia should be volcanic, that is the part. It is only in this part of our island continent that we find extinct craters, such as Mount Gambier in South Australia, and many undoubted volcanic vents in Victoria, while much of New South Wales is also of undoubted volcanic origin, the cause of which can only be sought in the action of these two “deeps,” one against the other.

The Abrolhos group extends for a distance of about sixty miles north and south, and if they had been upheaved by any such action as that above referred to, the area on which they stand must have been broken up into ridges and contorted, and its original levels, therefore, must have been altered relatively one to the other, while portions which were not of coral formation would also have been forced above water. As these islands, however, are all of that formation, and all of nearly uniform elevation, the evidence is pretty conclusive that upheaval has had no share in the cause of their appearance.

But subsidence of the ocean bed has another and different effect, which causes the appearance of dry land without upheaval. The world is continually shrinking in diameter at the rate of about four feet in a year. The sun’s present rate of contraction, as determined by astronomers, is four miles in a century, that of the earth is one mile in about fourteen centuries.

Every volcanic outburst, every earth tremor, is the effect of this shrinking of the earth’s diameter, and there is no outpouring of volcanic production that is not accompanied by a compensating subsidence at some part of the earth’s surface. But as the area of the oceans is two and a half times the area of dry land, it follows that five-sevenths of the subsidences which take place are in the ocean, and, therefore, not visible to us, while much of that which takes place even on the land is in inaccessible and uninhabited parts, of which we have no cognisance. Thus we know of but a very small proportion of the subsidences which are continually taking place, although the seismograph, an invention of recent years, places on record very many

earth tremors of which we otherwise would have no knowledge. All these tremors are the result of the settlement of the earth's crust, to accommodate itself to the continually altering strain of which volcanic outbursts are the visible effect.

These subsidences of the ocean bed, drain the waters off the shallower portions, or oceanic plateaux, such as that on which the Abrolhos stand. If the highest of these coral islands had been built up to the water level, and a series of subsidences in various parts of the ocean had taken place which lowered the surface of the waters, say, twelve feet, that would leave those islands standing twelve feet out of the water; while those portions of the plateau on which the coral insects had only built to within eight feet of the surface, would be left four feet out of the water, and the relative levels of the islands would thus remain unaltered.

One cannot sail amongst the Abrolhos group without being struck with the unaltered relative positions in level which they occupy, and, therefore, that their existence to-day is due to subsidences in the ocean's bed which have taken place in various parts of the world, probably at various periods, to tell us of seismic disturbances of which we otherwise have no record.

TRANSCRIBER'S NOTES:

Obvious typographical errors have been corrected.

The text on page 26 from *Pinafore* has been retained from the original, although the actual text of the second line is "And seek the seclusion that a cabin grants."

*** END OF THE PROJECT GUTENBERG EBOOK CHRISTMAS ON
THE BRINY ***

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